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Complexity of locations: more thrilling when close

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Problem: Why does the interest of events vary with the distance from home?

This is a fundamental law of journalism: greater distance tend to make identical events less interesting. Casualties in a tragic event are more newsworthy than distant ones. The phenomenon is known as the hierarchy of deaths, or as the "law of distant deaths".

Question: Does Algorithmic Information Theory have anything to say about the "hierarchy of deaths"?

Answer: Yes, definitely! It even predicts the phenomenon.

Complexity of Locations and Newsworthiness



as temporal landmarks.

And you use it also to designate

individuals among your acquaintances,

using common friends as landmarks.

Conclusion:

The interest of an event decreases

as twice the logarithm of the distance

to the observer;

but proximity to famous landmarks can

dramatically increase interest

► Complexity of Locations and Newsworthiness in a nutshell

On December 25, 2003, the French news reported three accidents in the following order:

1. a small plane crash in the French Alps (2 casualties),
2. another plane crash in Benin, Africa, (113 victims),
3. a gas plant explosion in China (230 victims).

Distance effects 1

0/1 point (ungraded)

Distances from Paris to these three events are 0.5 Mm, 4 Mm and 8 Mm (Mm = mega-metre). What is the difference in bits of complexity between these locations (B-A and C-A)?

☒ B-A: 3 bits; C-A: 4 bits

☐ B-A: 6 bits; C-A: 8 bits

☐ B-A: 8 bits; C-A: 16 bits



Answer

incorrect:
The distance ratios are 8 and 16, which correspond to 3 and 4 bits. Compared with the first event, greater distance adds 6 and 8 bits of complexity respectively ($2 \times \log_2 (dist.)$).

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You have used 1 of 1 attempt

 Answers are displayed within the problem

One may consider that dramatic impact depends on the logarithm of the number of casualties. We may write interest I as depending on distance d and on the number of casualties N :

$$I = - 2 \times \log_2(d) + b \times \log_2(1+N/a).$$

Parameter a provides the unit value on the stimulus axis. Here we consider that $a = 1$.
Parameter b measures the emotional interest due to the first casualty.
We ignore the effect of the cultural distance between the observer (here, the French audience) and the expected nationality of casualties. Cultural distance adds up to the cognitive complexity of the event, and thus contributes to diminishing its interest.

In the French news example, the most dramatic events were cited last: first A, then B and then C, though the ratio of casualties was 56 (B:A) and 115 (C:A). This means that the added complexity due to distance more than compensates for the emotion due to more numerous casualties.

Distance effects 2

1/1 point (ungraded)

What is the maximal value of coefficient b in the French news example?

☐ $b \leq 0.1$

☐ $b \leq 0.5$

☒ $b \leq 1$



Answer
Correct:
The ratio of casualties is 56 (B:A) and 115 (C:A). They translate into $b \log_{2+} (56) = 6 \times b$ bits and $b \log_{2+} (115) = 7 \text{ times } b$ bits respectively. With $b = 1$, shorter distances just compensate for the number of casualties.

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 Correct (1/1 point)

Measuring the distance of the event to the observer may not be the most economic way to measure description complexity. Quite often, events can be located in the vicinity of a known landmark (such as the Eiffel tower in Paris). The chain rule then applies to get an upper bound of $C_d(s)$ using landmark L :

$$C_d(s) \leq C_d(L) + 2 \times \log_2(\text{distance}(L, s))$$

Landmarks

1/1 point (ungraded)

Suppose that a dramatic event takes place in a foreign country. The media want to locate the event for the audience. A list of landmarks $\{L_i\}$ (cities, monuments, points of interest, etc.) is available. It is ranked by decreasing popularity. The event is located at distance d_i of landmark L_i . Which is the best way to locate the event?

☐ $i = 4$ and $d_i = 114 \text{ km}$



Correct:

$\text{argmin} (C(L_i) + 2 \times \log_2 (\text{distance}(L_i, x))),$
where $C(L_i)$ is assessed by $\log_2(i)$.

We "pay" 4 additional bits to go from $\log_{2+}(4) = 2$ for landmark $i = 4$ to $\log_{2+}(52) = 6$ for landmark $i = 52$, but we spare $2 \times \log_{2+}(114/15) = 6$ bits on the distance. So the second, less conspicuous landmark will be preferred.

You have used 1 of 1 attempt

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